Metabolic engineering: the study module

Part 1 — Biology foundations for engineers (weeks 1 to 10)

This is the reading, written out, so that you can work through weeks 1 to 10 without opening four books in parallel. It is written for someone who already has thermodynamics, kinetics, mass balances and transport, and who needs the biology those tools will be applied to. Every chapter maps onto one week of the tracker.

What this is not: it is not a substitute for Lehninger or Alberts, and it does not contain their text, their figures or their problem sets, which are copyrighted. I wrote every word here myself. Where their treatment is genuinely better than anything I can compress into a few pages — the three-dimensional structure figures, the full mechanisms, the pathway maps you will want to annotate — the chapter says so and names the chapter to open. Use this module as the spine and the books as the reference.

How to use it: read the chapter, then do that week's sessions in the tracker. The self-check questions at the end of each chapter are there to be answered on paper before you look at the answer. If you cannot answer them, reread rather than continuing: phase 1 compounds, and weeks 13 onward are unusable without it.

Ch.	Week	Chapter
1	1	The cell, the central dogma, and a map of the territory
2	2	Protein structure and enzyme function
3	3	Enzyme kinetics, in your own language
4	4	Bioenergetics and cofactors: the currency of the cell
5	5	Central carbon metabolism I: glycolysis and the pentose phosphate pathway
6	6	Central carbon metabolism II: the TCA cycle, respiration and overflow
7	7	Biosynthetic metabolism: where products come from
8	8	Molecular biology and gene regulation
9	9	Regulation at the system level: what will fight your design
10	10	Microbial genetics and microbiology for engineers

Chapter 1 · tracker week 1

The cell, the central dogma, and a map of the territory

> *A cell is a chemical reactor that builds its own catalysts, controls its own feed rates and makes copies of itself. Everything in this field follows from taking that description literally.*

What kind of reactor you are dealing with

A bacterial cell is about 1 µm across and holds roughly 1 femtolitre of cytoplasm — measured values for E. coli range from about 0.4 to 1.8 µm³, so take 1 fL as the round number and the range as the honest one. At a dense culture of 10^9 cells per millilitre, all that cytoplasm together is about 0.1% of the broth volume. That single number explains a great deal: a product kept inside the cell is capped at a fraction of a gram per litre however hard you push it, while a secreted product is limited only by what the cells can make and what the medium tolerates.

Inside, the cytoplasm is not dilute. Measured cytoplasmic protein in E. coli is 200 to 320 g/L, rising with external osmolarity, with total macromolecules near 300 g/L, and the whole interior is crowded enough that diffusion is slower than in water and weak associations between molecules are favoured. When you later read that an enzyme has a Km of 0.1 mM, remember that the cell is operating in a medium nothing like the assay buffer it was measured in.

Two architectures matter for production. Prokaryotes (bacteria, and the archaea you will mostly not use) have no internal membrane-bound compartments: one cytoplasm, one membrane in Gram-positives, two in Gram-negatives with a periplasm between them. Eukaryotes (yeasts, algae, mammalian cells) partition their chemistry into a nucleus, mitochondria, an endoplasmic reticulum and Golgi for the secretory route, peroxisomes and a vacuole. Compartments are both an opportunity and a tax, and week 58 of the tracker returns to this with numbers.

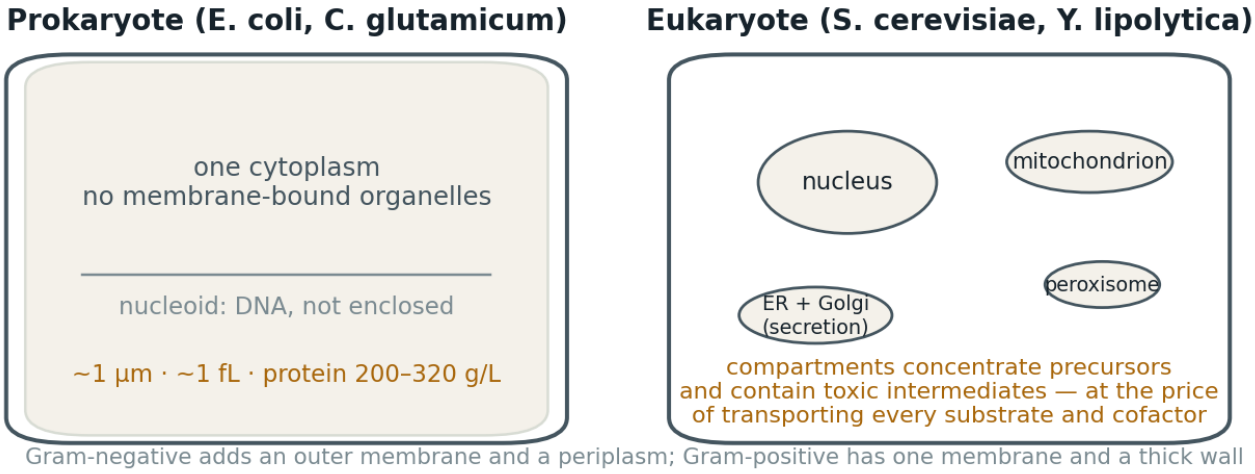


Figure 1.1 Two architectures. Everything in the module that depends on compartments, secretion or transport traces back to this picture. Drawn for this module.

The central dogma, stated as an engineer needs it

DNA is copied to DNA (replication), transcribed to RNA, and RNA is translated to protein. The direction matters because it is where your control points are. You can change the gene sequence, the amount of mRNA made from it, how efficiently that mRNA is translated, and how long the protein survives. Each of these is a separate engineering handle with a different cost and a different response time.

Three numbers worth carrying. Transcription in E. coli runs at roughly 40 to 80 nucleotides per second and translation at 10 to 20 amino acids per second, so a 1,000-codon gene takes on the order of a minute to express. Most bacterial mRNAs have half-lives of a few minutes, which is why transcriptional control acts quickly. Most proteins are stable over a cell generation, so once a protein is made, turning off its gene removes it mainly by dilution as the cells grow. That last point catches people out: a knockdown at the RNA level does not clear the protein, and a fast-growing culture dilutes it faster than a slow one.

The vocabulary you actually need

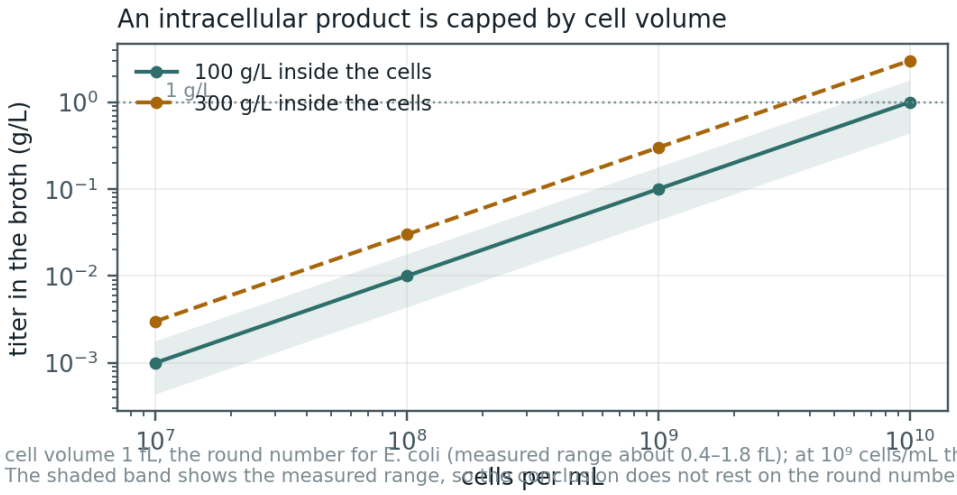
A gene is a stretch of DNA encoding a product. An open reading frame is the part translated into protein, read in codons of three bases. A promoter is the DNA where RNA polymerase binds and starts transcription; its strength sets how much mRNA is made. A ribosome binding site, in bacteria, is the sequence where the ribosome engages, and it sets translation initiation rate. A terminator ends transcription. An operon is several genes transcribed from one promoter as a single mRNA.

A plasmid is a small circular DNA molecule that replicates independently of the chromosome; its copy number, from a handful to hundreds per cell, multiplies the dose of whatever it carries. An integrated construct sits in the chromosome at one copy, which is more stable and usually lower expression. Choosing between them is a real design decision and reappears in week 20.

Where the carbon goes

A heterotroph takes in a reduced carbon source, oxidises part of it to make energy and reducing power, and uses the rest as building blocks. Three fates compete for every carbon atom you feed: biomass, product, and CO_2. Nothing you do in metabolic engineering escapes that three-way split, and your yield is a statement about it.

Biomass is not free. Roughly half of dry cell weight is protein, with nucleic acids, lipid and cell wall making up most of the rest, and building all of it consumes carbon, ATP and reducing power. This is why a strain that grows beautifully often produces poorly: the two demands draw on the same pool. The tension is quantified in weeks 18 and 19.



cell volume 1 fL, the round number for E. coli (measured range about 0.4–1.8 fL); at 10^9 cells/mL the cytoplasm is 0.1% of the broth. The shaded band shows the measured range, so the conclusion does not rest on the round number.

Figure 1.2 Why an intracellular product is capped. Computed from a 1 fL cell volume with the measured E. coli range shaded; the dotted line is 1 g/L, which most processes must clear to be worth running.

The equations from this chapter

Cell volume fraction	$\varphi_{cell} = N \cdot V_{cell}$, so 10^9 cells/mL × 1 fL = 0.001 L per L of broth
Carbon split	$C_{substrate} = C_{biomass} + C_{product} + C_{CO_2}$, always

Worked example

A culture reaches 10^9 cells/mL. Your product is retained in the cytoplasm at 150 g/L inside the cells. What titer do you measure in the broth?

The intracellular volume is 10^9 cells/mL × 10^3 mL/L × 1×10^{-15} L = 0.001 L per litre of broth, so 0.1%. The broth titer is 150 g/L × 0.001 = 0.15 g/L, and even at the top of the measured cell-volume range, 1.8 fL, it is only 0.27 g/L. Even an implausibly high intracellular concentration gives a titer that would be dismissed as a failure. This is the single strongest argument for secretion, and it is why hosts are chosen partly on whether they export.

Three things people get wrong

- ‘The cell is a bag of dilute enzymes.’ It is not: 200 to 320 g/L of protein, crowded, with slower diffusion than water.
- ‘Turning off the gene removes the protein.’ Mostly it does not; growth dilutes it.
- ‘More plasmid copies is always more product.’ More copies also means more burden and more instability.

Self-check — answer on paper first

1. Why does a secreted product have no cell-volume ceiling, and an intracellular one does?

Because the intracellular product occupies only the cytoplasmic volume, which is about 0.1% of the broth at 10^9 cells/mL, so the broth concentration is the intracellular concentration times that fraction. A secreted product occupies the whole liquid volume, so its ceiling is set by solubility, toxicity and what the cells can make, not by cell volume.

2. Name the four levels at which you can set how much of a protein a cell contains, and the timescale of each.

Gene sequence and copy number (permanent, set at construction); transcription rate via the promoter (minutes, because mRNA turns over in minutes); translation initiation via the ribosome binding site (minutes); protein degradation via a degradation tag (otherwise slow, since most proteins are cleared mainly by dilution through growth).

3. Why is a strain that grows well often a poor producer?

Because biomass and product draw on the same carbon, ATP and reducing power. Carbon has three fates: biomass, product and CO_2. Fast growth claims the first, so unless the pathway is decoupled from growth or the process is staged, the two compete directly.

What to open a book for

Alberts Ch. 1–2 for the cell and its compartments, and Lehninger Ch. 1 for the chemical logic, are worth reading for their figures rather than their text. If you buy only one thing for this week, buy nothing: MIT OCW 7.05 lectures 1 and 2 cover it free.

Chapter 2 · tracker week 2

Protein structure and enzyme function

An enzyme is a catalyst whose selectivity comes from a folded shape. Everything you will later do to enzymes — engineering them, expressing them, failing to express them — depends on that shape being fragile and context-dependent.

Four levels, and why each one can ruin your day

Primary structure is the amino acid sequence, written from the gene. Secondary structure is the local pattern the backbone takes, chiefly α-helices and β-sheets, held by hydrogen bonds along the backbone. Tertiary structure is the three-dimensional fold of one chain, held by the burial of hydrophobic side chains, hydrogen bonds, salt bridges and sometimes disulfide bonds. Quaternary structure is the assembly of several chains into one functional unit.

For your purposes: primary structure is what you design and order. Tertiary structure is what fails when you express a plant enzyme in E. coli at 37 °C and get an inactive pellet. Quaternary structure is why some enzymes lose activity when you tag them at the wrong terminus, and why an enzyme can be present on a gel and dead in an assay.

The twenty side chains, grouped usefully

You do not need to memorise structures, but you do need the groups, because they predict where a residue will sit and what a mutation will do. Non-polar side chains (glycine, alanine, valine, leucine, isoleucine, methionine, proline, phenylalanine, tryptophan) are mostly buried in the core, and substituting a buried large residue for a small one leaves a cavity and destabilises the fold. Polar uncharged side chains (serine, threonine, cysteine, tyrosine, asparagine, glutamine) sit at the surface or in active sites; serine, threonine, tyrosine and cysteine are the usual nucleophiles and the usual sites of modification. Acidic (aspartate, glutamate) and basic (lysine, arginine, histidine) side chains sit at the surface and form salt bridges; histidine is special because its pKa near 6 means it can be protonated or not at physiological pH, which is why it appears so often in catalysis and in metal binding.

Two practical consequences. First, cysteine is reactive: in the reducing cytoplasm it stays free, so a protein that needs disulfide bonds will not fold properly there, which is one reason secreted and extracytoplasmic proteins often need a different compartment or host. Second, proline breaks helices and constrains the backbone, so a proline substitution in a helix is rarely neutral.

What catalysis actually is

An enzyme does not change the equilibrium of a reaction. It lowers the activation barrier for both directions equally, so it changes the rate, not the position. If you remember one thing from this week, remember that: no amount of enzyme engineering makes an unfavourable reaction favourable, and a pathway step that is thermodynamically stuck stays stuck. Week 14 gives you the arithmetic for deciding whether a step is stuck.

How the barrier is lowered: by binding the transition state more tightly than the substrate, by positioning reacting groups correctly so that the entropic cost of getting them together is paid once at binding, by providing acid and base groups to move protons, by covalent catalysis through a nucleophilic residue, and by metal ions that stabilise charge. The common thread is that the active site is a pocket complementary to the transition state, not to the substrate.

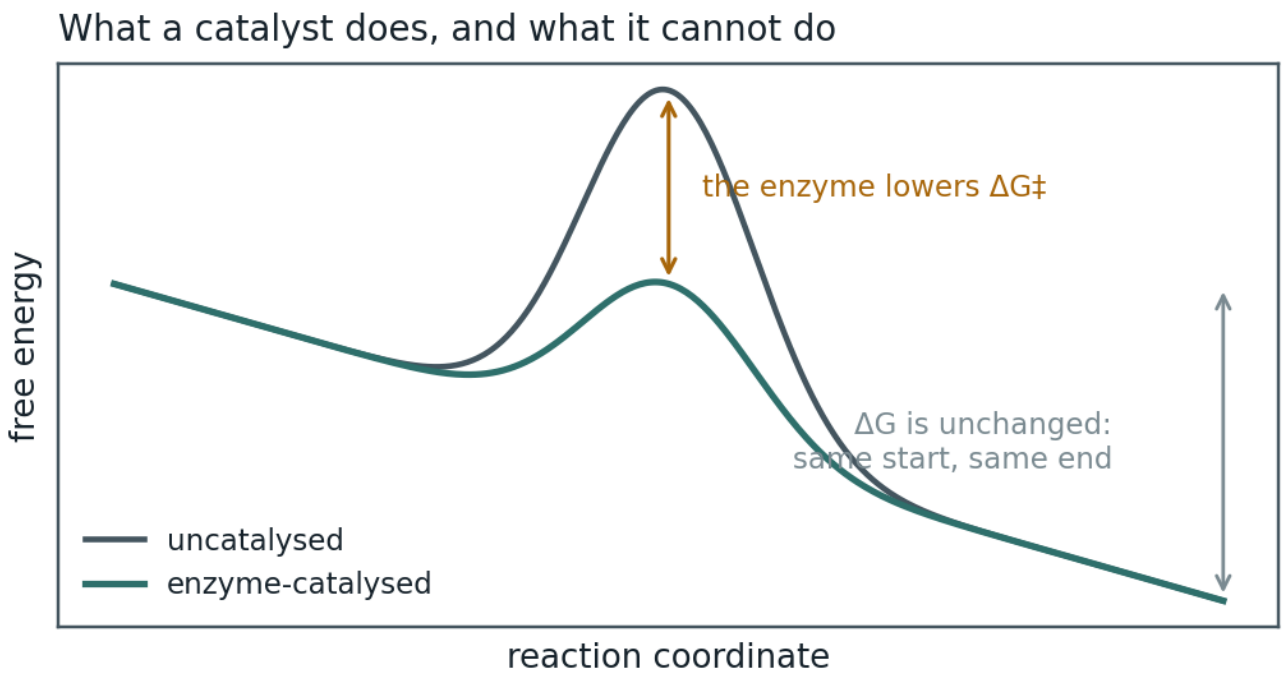


Figure 2.1 A catalyst lowers the activation barrier in both directions and leaves ΔG untouched. If you remember one diagram from week 2, this is the one.

Cofactors, because your pathway will need them

Many enzymes cannot work alone. Cofactors divide into metal ions (magnesium with nearly every phosphoryl transfer, iron in haem and iron-sulfur clusters, zinc, manganese, copper) and organic cofactors, which are either tightly bound prosthetic groups or freely diffusing cosubstrates. The ones you will meet constantly are NAD^+ and $NADP^+$ as hydride carriers, ATP as the phosphoryl donor, coenzyme A as the acyl carrier, and thiamine pyrophosphate, pyridoxal phosphate, biotin and tetrahydrofolate for specific chemistry.

The engineering point is that a cosubstrate must be regenerated. When you install a pathway that consumes NADPH, you have not just added a reaction, you have added a demand on the cell's reductant supply, and that demand competes with everything else. Week 4 makes this quantitative and week 19 makes it a design constraint.

The equations from this chapter

Catalysis	`k_cat/K_M is the specificity constant; an enzyme lowers ΔG‡, never ΔG`
Why histidine	`pKa ≈ 6 means it is a usable acid or base at pH 7`

Worked example

You express a plant enzyme in E. coli. It appears as a strong band at the right size, but there is no activity in the lysate. Give the three hypotheses you would test first, with the test for each.

First, misfolding into inclusion bodies: run soluble and insoluble fractions side by side on a gel, since a strong total band says nothing about which fraction it is in. Second, a missing cofactor or metal, or a missing redox partner, which is common for plant cytochrome P450 enzymes: add the cofactor to the assay, or co-express the partner. Third, incorrect disulfide bonding or a missing post-translational modification, since the E. coli cytoplasm is reducing: test expression in a compartment or host that permits it. A fourth, often overlooked, is that the assay conditions are wrong for this enzyme: vary pH and temperature before concluding the protein is dead.

Three things people get wrong

- ‘A band on a gel means the enzyme works.’ It means something of that size is present.
- ‘Enzymes make reactions go.’ They make reactions go faster, in both directions.
- ‘Codon-optimise everything.’ Translation that is too fast can fold worse; this is a hypothesis to test, not a rule.

Self-check — answer on paper first

1. Why can an enzyme not make a thermodynamically unfavourable reaction proceed?

Because a catalyst lowers the activation barrier for the forward and reverse directions equally, leaving the equilibrium constant unchanged. What determines direction is $\Delta G'$ at the actual concentrations, so the only way to drive an unfavourable step is to change the concentrations, by removing the product or coupling the step to a favourable reaction.

2. Why does the reducing cytoplasm of E. coli matter when you express a eukaryotic secreted protein?

Because such proteins often depend on disulfide bonds for their fold, and the cytoplasm keeps cysteines reduced, so the bonds do not form and the protein misfolds. The usual answers are to target the protein to the periplasm, to use a strain engineered with an oxidising cytoplasm, or to use a eukaryotic host with an endoplasmic reticulum.

3. Name the cofactor role you would check first for a pathway consuming four NADPH per product, and why.

Whether the host can supply NADPH at the required rate, because NADPH is a cosubstrate that must be regenerated, usually through the oxidative pentose phosphate pathway, which costs carbon as CO_2. A pathway with a large NADPH demand is a demand on central metabolism, not just an addition to it.

What to open a book for

This is the week where the books earn their price: Lehninger Ch. 3 to 6 for amino acids, protein structure and enzymes, and the structure figures in Alberts. No prose substitute shows you a fold. Cornish-Bowden is the reference if you want the kinetics done properly.

Chapter 3 · tracker week 3

Enzyme kinetics, in your own language

You already know rate laws, steady-state approximations and reactor design. This week is mostly a translation exercise, plus one piece of biology that has no chemical-engineering analogue: allosteric regulation.

Michaelis–Menten, derived the way you would derive it

Take E + S ⇌ ES → E + P. Apply the steady-state approximation to ES, exactly as you would to a reactive intermediate, and impose the enzyme balance $E_{total} = E + ES$. The result is $v = V_{max}\cdot[S]/(K_M + [S])$, with $V_{max} = k_{cat}\cdot E_{total}$ and $K_M = (k_{-1} + k_{cat})/k_1$. That is the whole derivation, and it is the same algebra as a Langmuir–Hinshelwood rate law on a catalyst surface; the enzyme is simply a one-site catalyst.

Read the two constants physically. V_{max} is capacity: it scales with how much enzyme you have, so doubling expression doubles it. K_M is the substrate concentration giving half of V_{max}, and it tells you where on the curve the cell operates. If the intracellular substrate concentration is well above K_M the enzyme is saturated and adding substrate does nothing; if it is well below, the rate is roughly proportional to substrate and the enzyme behaves first order with apparent rate constant $k_{cat}\cdot E/K_M$.

What the constants do and do not tell you

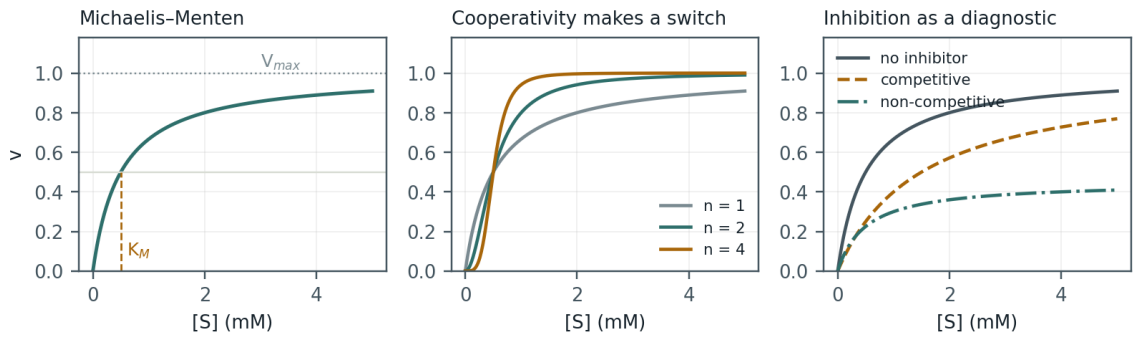
k_{cat}/K_M, the specificity constant, is the figure of merit for an enzyme working at low substrate, and it is bounded above by the rate at which substrate and enzyme can meet by diffusion, around 10^8 to 10^9 $M^{-1}\,s^{-1}$. An enzyme at that bound cannot be improved by protein engineering; it can only be given more substrate or more copies.

A warning you will need in week 31. A higher k_{cat} measured in a cuvette often does not raise titer in a cell, because the enzyme may not be the step that controls flux, because expression or solubility may fall as activity rises, or because the substrate concentration in the cell sits in a different regime from the assay. Kinetic constants are necessary, not sufficient.

Inhibition, read as a diagnostic

Competitive inhibition raises the apparent K_M and leaves V_{max} unchanged: the inhibitor competes for the site, and enough substrate outruns it. Uncompetitive inhibition lowers both, binding only the enzyme-substrate complex. Non-competitive or mixed inhibition lowers V_{max}, because the inhibitor binds elsewhere and reduces catalysis whatever the substrate does. The practical value of these patterns is diagnostic: measure the rate at several substrate and inhibitor concentrations, see which constant moves, and you know where the inhibitor acts.

Product inhibition is the case you will meet most often in a fermentation, and it is usually competitive or mixed. It is also the reason that the two-phase and in situ removal strategies of weeks 44 to 47 exist.



competitive inhibition raises apparent K_M and leaves V_{max} alone; non-competitive lowers V_{max}. Measure which constant moves and you know where the inhibitor binds.

Figure 3.1 The three curves you must be able to read: the hyperbola, the sigmoid that makes a switch, and the two inhibition patterns that tell you where an inhibitor binds.

Allosteric regulation, the part that is not chemical engineering

Many regulatory enzymes do not follow Michaelis–Menten at all. They have several subunits, and substrate binding at one site changes the affinity at the others, giving a sigmoidal rather than hyperbolic curve. The standard empirical description is the Hill equation, v = V_max·[S]ⁿ/(Kⁿ + [S]ⁿ), where n is a measure of cooperativity rather than a number of sites.

Why it matters: a sigmoidal response is a switch. A small change in substrate or effector concentration produces a large change in rate, which lets a cell respond sharply to a threshold. Feedback inhibition, where the end product of a pathway inhibits the first committed enzyme allosterically, is the most common regulatory motif you will fight. The classic engineering answer is to use a feedback-resistant variant of that enzyme, and you will see this in nearly every amino-acid production paper you read.

The equations from this chapter

Michaelis–Menten	`v = V_max·[S] / (K_M + [S]), V_max = k_cat·[E]_total`
Hill	`v = V_max·[S]ⁿ / (Kⁿ + [S]ⁿ), n > 1 means cooperative`
Diffusion bound	`k_cat/K_M ≤ about 10⁸ to 10⁹ M⁻¹ s⁻¹`

Worked example

An enzyme has k_cat = 50 s⁻¹ and K_M = 0.5 mM. The intracellular substrate concentration is 50 µM. What fraction of V_max is the enzyme achieving, and what would you do to raise flux through this step?

v/V_max = [S]/(K_M + [S]) = 0.05/(0.5 + 0.05) = 0.091, so about 9% of capacity. The enzyme is far from saturated and is operating in its first-order regime, so flux here is set by substrate supply as much as by enzyme amount. Options, in order of likely payoff: raise the intracellular substrate concentration by improving the upstream step or reducing a competing drain; express more enzyme, which raises V_max proportionally and therefore raises the first-order rate too; or engineer a lower K_M, which is the hardest route. Before any of them, check in week 15 whether this step actually controls the flux.

Three things people get wrong

- ‘K_M is the enzyme’s affinity.’ It is a composite of three rate constants and only approximates affinity when k_cat is small.
- ‘A better k_cat means more product.’ Only if that enzyme controls the flux and expression holds up.

- ‘Sigmoidal curves are just noisy hyperbolas.’ They are the signature of cooperativity, and they are how cells build switches.

Self-check — answer on paper first

1. Derive, in one line each, why V_max scales with enzyme concentration but K_M does not.

V_max = k_cat·E_total comes from the enzyme balance, so it is proportional to how much enzyme is present. K_M = (k_{-1} + k_cat)/k_1 is a ratio of rate constants belonging to a single enzyme molecule, so it is a property of the enzyme and conditions, not of its amount.

2. You measure that an inhibitor raises the apparent K_M but leaves V_max unchanged. What does this tell you, and what follows for a process?

It is competitive inhibition, so the inhibitor binds the free enzyme at the substrate site. It follows that raising the substrate concentration can overcome it, which in a process means the inhibition matters most when substrate is limiting, as in a fed-batch running at low residual sugar.

3. Why is a feedback-resistant enzyme variant such a common feature of amino-acid production strains?

Because the end product allosterically inhibits the first committed enzyme of its own pathway, so a strain that accumulates the product switches off its own synthesis. A variant that no longer binds the product at its allosteric site keeps the pathway on, which is usually the single most effective change in such a strain, at the cost of losing the regulation that protected the cell.

What to open a book for

Lehninger Ch. 6 covers this, but you will find Cornish-Bowden or the enzyme-kinetics chapter of Shuler, Kargi & DeLisa closer to how you think. Shuler in particular derives the same equations in engineering notation and connects them straight to reactor design.

Chapter 4 · tracker week 4

Bioenergetics and cofactors: the currency of the cell

This is the most important week in phase 1 for what comes later. Almost every design failure in metabolic engineering is, underneath, an energy or redox balance that does not close.

ΔG, and why the textbook number misleads

Tabulated standard free energies, ΔG°′, assume 1 M concentrations of everything, pH 7 and 25 °C. A cell operates at micromolar to millimolar concentrations. What decides direction is ΔG′ = ΔG°′ + RT·ln Q, where Q is the actual mass-action ratio of products to substrates. The consequence is that a reaction with an unfavourable standard free energy runs perfectly well if its product is kept low, and a favourable one stalls if its product accumulates.

This single idea does more work than any other in the field. It is why removing a product in situ can pull a pathway forward; it is why an intermediate that accumulates tells you where a pathway is stuck; and it is why you should never declare a designed pathway infeasible from standard free energies alone. Week 14 gives you the full calculation with real concentrations.

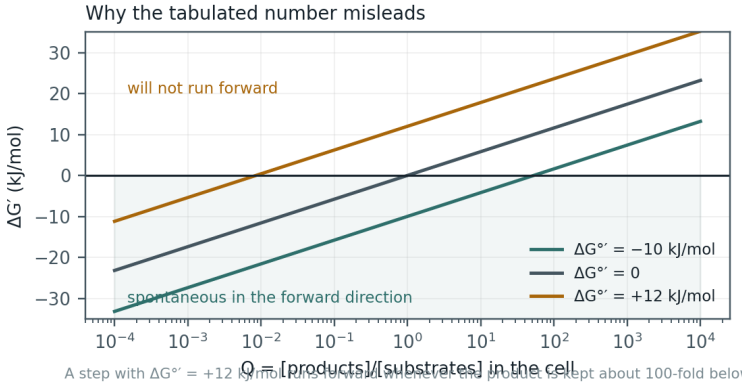


Figure 4.1 ΔG′ against the mass-action ratio, computed at 30 °C. The shaded region is where a reaction runs forward. This is the argument for in situ product removal, drawn.

ATP, properly understood

ATP is not a store of energy in any useful sense; the cell holds only seconds to a minute of demand in its ATP pool. It is a carrier, constantly regenerated. What matters is the rate of regeneration and the ratio ATP to ADP and phosphate, which sets the free energy actually available per mole transferred. Compilations give roughly 40 to 60 kJ/mol in a living cell against the 30.5 kJ/mol standard value, and a measurement in growing E. coli put it near 47 kJ/mol. Quote a range, not a number.

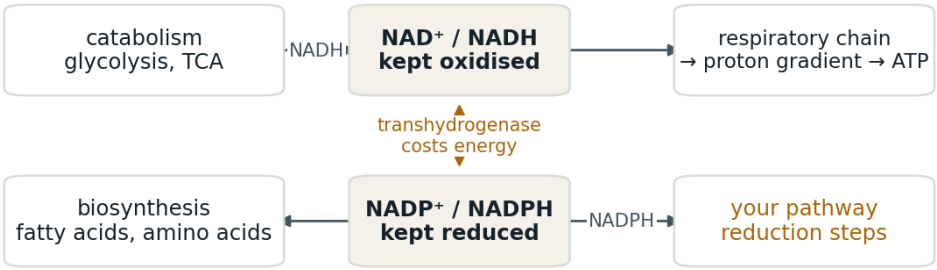
Count ATP in equivalents, not molecules. Hydrolysis to ADP is one equivalent; hydrolysis to AMP, as in acyl-CoA activation, is two, because regenerating ATP from AMP costs two phosphoryl transfers. Getting this wrong by a factor of two in a pathway balance is a common and consequential mistake.

NADH and NADPH are not interchangeable

Both carry a hydride. The difference is a single phosphate group, and the cell uses it as a label: NADH is kept relatively oxidised and is the carrier for catabolism, feeding the respiratory

chain to make ATP, while NADPH is kept relatively reduced and is the carrier for biosynthesis. The ratios differ by orders of magnitude, which is the point: it lets the cell run oxidation and reduction at the same time without futile cycling.

For you this means that a pathway requiring NADPH cannot simply draw on the NADH your host makes abundantly. The options are a transhydrogenase, which converts one to the other at an energetic cost; routing flux through the oxidative pentose phosphate pathway, which generates NADPH but loses carbon as CO_2; or engineering the enzyme's cofactor specificity. Each appears in real strain papers, and each has a cost you can quantify.



Same hydride, different label. The cell holds the two pools at deliberately different ratios so that it can oxidise and reduce at once.

Figure 4.2 Two hydride carriers, deliberately held at different ratios. A pathway needing NADPH cannot simply draw on the NADH the host makes.

How respiration actually makes ATP

Oxidation of NADH by the respiratory chain pumps protons across the membrane, creating an electrochemical gradient — the proton motive force — and ATP synthase lets protons back in, turning that gradient into ATP. The yield is not a fixed integer: it depends on which dehydrogenases and oxidases the organism uses, and the P/O ratio differs between organisms and conditions. Treat any number you are given for ATP per NADH as a modelling assumption to be stated, not a constant.

The proton motive force also does other work: it drives many transporters, including the efflux pumps of week 44, and it powers flagella. When you export a product using a proton antiporter, you are spending the same gradient that would otherwise have made ATP.

The equations from this chapter

Real free energy	`ΔG′ = ΔG°′ + RT·ln Q; R = 8.314 J/mol/K, RT ≈ 2.5 kJ/mol at 30 °C`
ATP equivalents	`ATP → ADP + Pi = 1 equivalent; ATP → AMP + PPi = 2 equivalents`
Redox balance	`sum of NAD(P)H produced = sum consumed, per unit of product, or the strain cannot run`

Worked example

A pathway makes one mole of product from one mole of glucose, consuming 4 NADPH and 2 ATP, and releasing 2 CO_2. Glycolysis to two pyruvate yields 2 NADH and 2 ATP net. Is this balanced, and what follows?

Carbon: glucose has 6, two CO_2 leave, so 4 carbons reach the product; check that against the product's formula. ATP: glycolysis supplies 2, the pathway spends 2, so ATP is break-even before any maintenance, which means there is nothing left for growth — a red flag. Redox: glycolysis supplies 2 NADH, the pathway needs 4 NADPH, so the strain is short by 4 NADPH and has 2 NADH it does not need. Nothing about this balances by itself. The realistic resolutions are to route some glucose through the oxidative pentose phosphate pathway to make NADPH at a cost in carbon, to install a transhydrogenase at a cost in energy, or to change the pathway enzymes' cofactor preference to NADH. Which you choose is a quantitative question, and week 19 answers it.

Three things people get wrong

- ‘ATP stores energy for the cell.’ It carries it; the pool lasts seconds.
- ‘NADH and NADPH are the same thing.’ They are kept at deliberately different ratios and serve opposite purposes.
- ‘Standard free energy tells me whether my pathway works.’ Only $\Delta G'$ at real concentrations does.

Self-check — answer on paper first

1. Explain, using the equation, how in situ product removal can make a pathway step feasible.

$\Delta G' = \Delta G^{\circ\prime} + RT \ln Q$, and Q contains the product concentration in the numerator. Removing the product lowers Q, which lowers $\Delta G'$ and can turn a positive value negative. The step itself is unchanged; only the concentrations are. This is the thermodynamic half of the argument for two-phase fermentation, the other half being relief of toxicity.

2. Why does activating a fatty acid as an acyl-CoA cost two ATP equivalents rather than one?

Because the reaction cleaves ATP to AMP and pyrophosphate rather than to ADP and phosphate. Regenerating ATP from AMP requires two phosphoryl transfers, so the true cost is two equivalents. Counting it as one understates the energy demand of every acyl-CoA-dependent pathway, which includes polyketides and fatty acid products.

3. A colleague gives you a model that assumes 3 ATP per NADH oxidised. What is your response?

That it is an assumption rather than a constant: the ATP yield depends on which respiratory complexes the organism uses, the proton stoichiometry of its ATP synthase, and the conditions, and the proton motive force is also spent on transport. The right response is to state it as an assumption, check how sensitive the conclusion is to it, and look for a measured P/O value for that organism.

What to open a book for

Lehninger Ch. 13 is the chapter to read, and it is worth reading twice. Doran's material- and energy-balance chapters cover the accounting side in engineering notation, with problems you can actually work.

Central carbon metabolism I: glycolysis and the pentose phosphate pathway

Central metabolism is the feedstock supply system for every pathway you will install. You need to know it well enough to say, from memory, where a precursor comes from and what making it costs.

Glycolysis as a two-stage process

The first stage invests: glucose is phosphorylated and split, consuming 2 ATP, to give two molecules of glyceraldehyde-3-phosphate. The second stage pays back: each three-carbon unit is oxidised and dephosphorylated to pyruvate, yielding 2 ATP and 1 NADH per unit. Net per glucose: 2 ATP, 2 NADH, 2 pyruvate. Those three numbers are worth knowing cold, because almost every balance you write starts from them.

The intermediates matter more than the steps. Glucose-6-phosphate is the branch point to the pentose phosphate pathway. Fructose-6-phosphate and glyceraldehyde-3-phosphate are where the non-oxidative pentose phosphate reactions rejoin. Dihydroxyacetone phosphate is the precursor of glycerol and therefore of lipid backbones. Phosphoenolpyruvate and pyruvate are the precursors of aromatic amino acids, of the anaplerotic reactions, and of a large fraction of engineered products. When you read a pathway paper, it is these names, not the enzyme names, that tell you what is being competed for.

Alternative routes, which exist for a reason

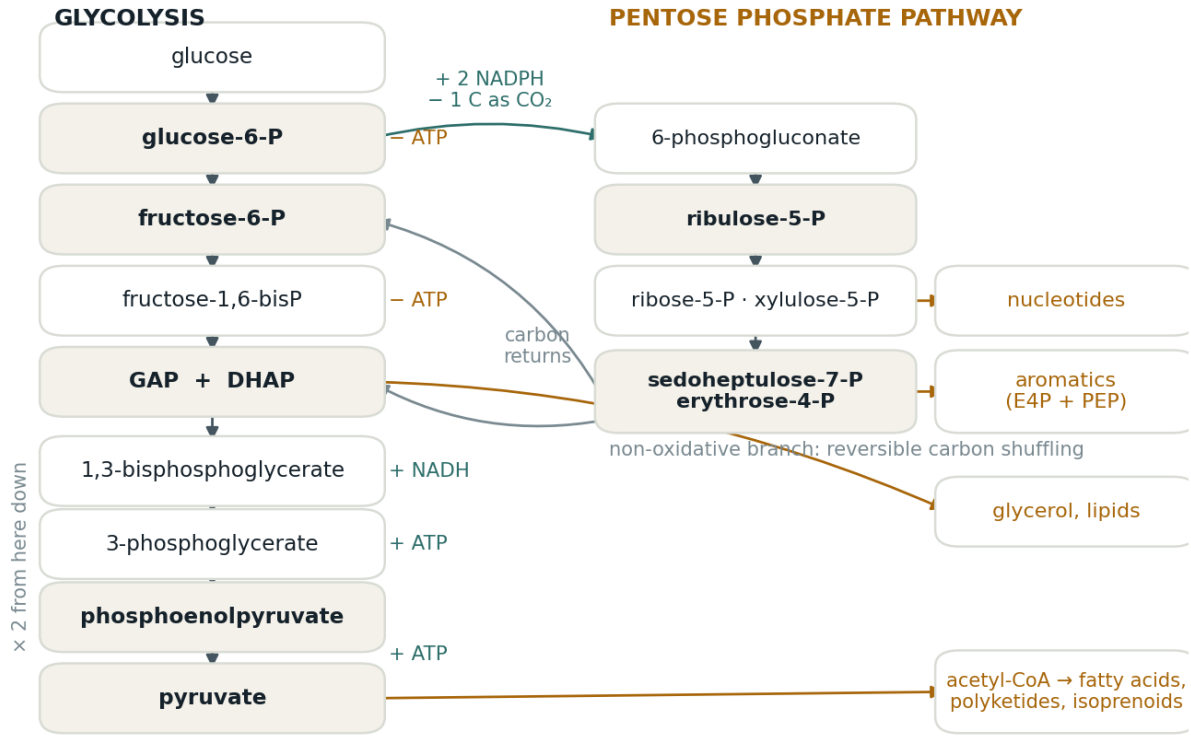
Not every organism uses the glycolysis you learned. The Entner–Doudoroff pathway, used by Pseudomonas and others, yields only 1 ATP per glucose but is calculated to need several-fold less enzymatic protein for the same glucose throughput, which is a sensible trade when carbon is plentiful and protein is expensive (Flamholz et al., PNAS 110:10039, 2013). The phosphoketolase pathway, found in some bacteria, splits sugars to give acetyl phosphate and avoids the two-carbon loss of pyruvate decarboxylation, which is why it appears in engineered strains aiming at high acetyl-CoA yield.

The lesson is not to memorise variants but to ask, for any host, which route it uses, because the ATP and redox yields differ and your balance depends on them.

The pentose phosphate pathway: the NADPH tap

The oxidative branch takes glucose-6-phosphate, oxidises it twice and decarboxylates it, producing 2 NADPH and one CO_2 per glucose entering, and leaving ribulose-5-phosphate. The non-oxidative branch is a set of reversible carbon-shuffling reactions, transketolase and transaldolase, which interconvert sugar phosphates of three to seven carbons and let the cell return carbon to glycolysis or draw ribose-5-phosphate for nucleotides and erythrose-4-phosphate for aromatic amino acids.

This is the pathway you will manipulate most often, because it is where NADPH comes from. The cost is explicit: every glucose sent through the oxidative branch for reductant loses a carbon as CO_2. That is the trade-off behind the lysine calculation in week 59 and behind the lipid ceiling in week 58. Flux through this branch is not fixed; it is regulated by NADPH demand, which is why strains with a high NADPH requirement naturally pull carbon this way, and why deleting phosphoglucose isomerase forces them to.



Net per glucose: 2 ATP, 2 NADH, 2 pyruvate.

Shaded boxes are the branch points every engineered pathway competes for. Drawn for this module, not traced from any source.

Figure 5.1 Glycolysis and the pentose phosphate pathway, with the branch points marked. The arithmetic of the NADPH-versus-carbon trade-off starts here.

The equations from this chapter

Glycolysis, net	glucose + 2 ADP + 2 Pi + 2 NAD^+ → 2 pyruvate + 2 ATP + 2 NADH + 2 H^+ + 2 H_2O
Oxidative PPP	glucose-6-P + 2 $NADP^+$ → ribulose-5-P + CO_2 + 2 NADPH
The trade	2 NADPH per glucose through the oxidative branch, at the cost of 1 carbon

Worked example

A product needs 4 NADPH per molecule and the only NADPH source is the oxidative pentose phosphate pathway. How much glucose does the reductant alone require per product molecule, and what does that do to the carbon yield?

Each glucose through the oxidative branch gives 2 NADPH, so 4 NADPH needs 2 glucose. Those 2 glucose each lose one carbon as CO_2 in the process, and their remaining five-carbon skeletons return to central metabolism rather than disappearing, so the carbon is not all lost — but the decarboxylation is. The practical consequence is that the carbon yield of such a product is always below the naive one-to-one bound, which is exactly what the verified lysine figure shows: a theoretical maximum of 0.81 mol per mol rather than 1.0, because reductant has to be paid for in carbon.

Three things people get wrong

- ‘Glycolysis makes a lot of ATP.’ It makes 2 net per glucose; respiration makes most of it.
- ‘The pentose phosphate pathway is for nucleotides.’ It is also the main NADPH source, which is usually why you care.

- ‘Every organism runs the same glycolysis.’ Entner–Doudoroff and phosphoketolase routes have different yields.

Self-check — answer on paper first

1. Name the five intermediates of central metabolism that most engineered pathways branch from, and one product from each.

Glucose-6-phosphate (sugar alcohols, and entry to the pentose phosphate pathway); fructose-6-phosphate and glyceraldehyde-3-phosphate (entry points for returning carbon, and precursors for amino sugars); dihydroxyacetone phosphate (glycerol and lipid backbones); phosphoenolpyruvate and erythrose-4-phosphate together (aromatic amino acids and aromatics); pyruvate and acetyl-CoA (fatty acids, polyketides, isoprenoids, organic acids). Credit any defensible list that names the branch point rather than the enzyme.

2. Why does a strain lacking phosphoglucose isomerase still grow on glucose, and what is its advantage?

Because glucose-6-phosphate cannot be isomerised into glycolysis, so carbon is forced through the oxidative pentose phosphate pathway and returns to lower glycolysis through the non-oxidative branch. It grows more slowly, because of the extra steps and the carbon lost as CO_2, but it generates far more NADPH, which is an advantage for an NADPH-demanding product.

3. Why is ATP per glucose a poor way to compare glycolytic routes for a production strain?

Because what you usually need is precursor supply and reductant, not ATP. A route with a lower ATP yield may be preferable if it avoids a decarboxylation, conserves carbon, or requires less enzyme protein. The right comparison is against the specific demands of your pathway, written as a balance.

What to open a book for

Lehninger Ch. 14 and 15. Read them with a blank sheet and redraw the pathway from memory afterwards; this is the one piece of memorisation in the whole curriculum that genuinely pays.

Chapter 6 · tracker week 6

Central carbon metabolism II: the TCA cycle, respiration and overflow

Where the rest of the carbon goes, where most of the ATP comes from, and why a well-fed culture excretes acetate.

The cycle, and its two jobs

Pyruvate is oxidised to acetyl-CoA, losing a carbon as CO_2 and reducing NAD^+. Acetyl-CoA then condenses with oxaloacetate and the resulting six-carbon acid is oxidised round to oxaloacetate again, releasing two CO_2 and generating three NADH, one reduced flavin and one nucleoside triphosphate per turn. Fully oxidising one glucose this way yields a large amount of reducing power, which the respiratory chain converts to ATP.

But the cycle has a second job that matters more for engineering: it is a supplier of precursors. α-Ketoglutarate is the nitrogen entry point and the precursor of glutamate and therefore of most amino nitrogen. Oxaloacetate is the precursor of aspartate and of the lysine and threonine families. Succinyl-CoA feeds haem synthesis. Drawing these off means the cycle loses carbon it needs to keep turning, which brings us to anaplerosis.

Anaplerosis, the word that will keep appearing

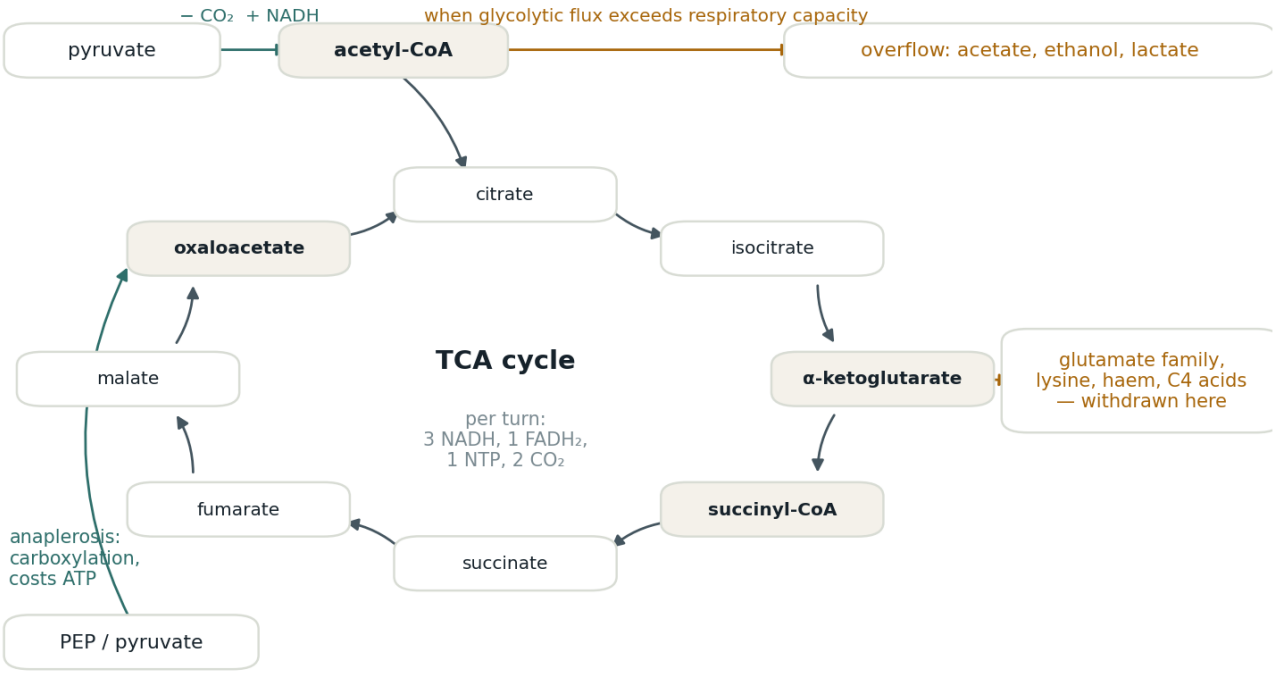
If you withdraw an intermediate, the cycle runs down, so cells have reactions that replenish them: carboxylation of pyruvate or phosphoenolpyruvate to oxaloacetate, and the glyoxylate shunt when growing on two-carbon substrates. These anaplerotic reactions are a frequent target in strain engineering, because a product drawn from oxaloacetate or α-ketoglutarate is limited by how fast the cell can refill the cycle, not by the pathway you installed.

This is one of the places where an engineer's instinct is right: it is a recycle loop with a purge, and the purge rate sets the make-up rate. If you are drawing product from the loop, you must feed it.

Overflow metabolism, and why your culture smells of vinegar

Feed E. coli plenty of glucose with plenty of oxygen and it still excretes acetate. This is not a defect, and the modern explanation is sharper than the older ‘limited respiratory capacity’ one. Making ATP by respiration costs more proteome per unit of energy than making it by fermentation, so above a threshold growth rate the cell gets more ATP per gram of protein invested by fermenting part of the carbon and excreting the product. Measurements in E. coli show acetate excretion rising linearly with growth rate above a threshold near 0.76 h^{-1}, and zero below it (Basan et al., Nature 528:99, 2015). The same logic produces ethanol in Crabtree-positive yeasts and lactate in mammalian cells. Either way the operational conclusion is the same: it is set by the specific uptake rate, so the feed rate controls it.

Three consequences for you. First, carbon you paid for leaves as a by-product, so yield falls. Second, acetate is itself inhibitory, so growth and production suffer at high concentrations. Third, the standard answer in practice is a fed-batch: feed slowly enough that glycolytic flux stays below the respiratory capacity. That is why nearly every industrial fermentation is fed-batch, and it is a process answer to a metabolic problem.



The cycle is a recycle loop with a purge: every intermediate you withdraw must be replaced, or the loop runs down.

Figure 6.1 The TCA cycle as a recycle loop with withdrawals, anaplerotic make-up, and the overflow outlet that appears when the feed outruns respiration.

The equations from this chapter

Pyruvate oxidation	`pyruvate + CoA + NAD⁺ → acetyl-CoA + CO₂ + NADH`
Per turn of the cycle	`3 NADH + 1 reduced flavin + 1 NTP + 2 CO₂`
Overflow condition	`excretion begins when glycolytic flux exceeds respiratory capacity`

Worked example

Your fed-batch accumulates 5 g/L acetate and growth has slowed. Your colleague proposes deleting the acetate pathway. What do you say?

That deleting the outlet does not remove the cause. Acetate appears because glycolytic flux exceeds respiratory capacity, so blocking the excretion route backs the pressure up into acetyl-CoA and the cycle, which commonly slows growth further or shifts the excretion to another by-product such as pyruvate or lactate. The first intervention is the feed: lower the feed rate so flux stays within respiratory capacity, which is the standard reason fed-batch exists. If a genetic change is wanted, raising respiratory or TCA capacity, or reducing glucose uptake rate, addresses the cause rather than the symptom. Deletion can work, but as part of a rebalanced strain, not on its own.

Three things people get wrong

- ‘Acetate means something went wrong.’ It means the feed exceeded respiratory capacity, which is normal and predictable.
- ‘The TCA cycle is for making ATP.’ It is also the precursor supply for nitrogen metabolism and several amino-acid families.
- ‘Deleting a by-product pathway raises yield.’ Often it moves the problem; fix the flux imbalance.

Self-check — answer on paper first

1. Why does withdrawing α-ketoglutarate for a product require an anaplerotic reaction?

Because the TCA cycle is a closed loop: every intermediate withdrawn is one fewer oxaloacetate returning to condense with acetyl-CoA, so the cycle runs down and both ATP generation and precursor supply fall. Anaplerotic carboxylation of pyruvate or phosphoenolpyruvate refills oxaloacetate, and the required anaplerotic flux is set by the withdrawal rate.

2. Explain why fed-batch operation is a metabolic answer and not merely a process convenience.

Because overflow metabolism is triggered by glycolytic flux exceeding respiratory capacity. By feeding substrate at a controlled rate, the fed-batch holds the specific uptake rate below the threshold, so carbon is fully oxidised or directed to product instead of being excreted as acetate or ethanol. It is a direct manipulation of the cell's flux distribution through the feed rate.

3. Name two products drawn from oxaloacetate and state what their yield depends on besides the installed pathway.

Aspartate-family products such as lysine, threonine and methionine, and C4 acids such as succinate and malate. Their yield depends on anaplerotic capacity, since oxaloacetate must be replenished by carboxylation, and that carboxylation costs ATP and requires a CO_2 supply; it also depends on competition from the cycle itself and from amino-acid biosynthesis.

What to open a book for

Lehninger Ch. 16 to 19. Alberts is weaker here. For overflow metabolism and proteome allocation, read a recent review rather than a textbook, since the quantitative account is recent.

Biosynthetic metabolism: where products come from

Twelve precursors from central metabolism build essentially everything. Knowing which one your target comes from tells you immediately where in metabolism you will be fighting.

The twelve precursors

Biosynthesis in a bacterium starts from about twelve compounds, all of them intermediates you met in weeks 5 and 6: glucose-6-phosphate, fructose-6-phosphate, ribose-5-phosphate, erythrose-4-phosphate, glyceraldehyde-3-phosphate, 3-phosphoglycerate, phosphoenolpyruvate, pyruvate, acetyl-CoA, α-ketoglutarate, succinyl-CoA and oxaloacetate. Every amino acid, nucleotide, lipid and cofactor is assembled from these, with ATP and reducing power supplied separately.

Treat this as a lookup table rather than a memory exercise. When you choose a target molecule, the first question is which precursor it derives from, because that determines which part of central metabolism must be amplified, what competes with you, and what the theoretical yield can be.

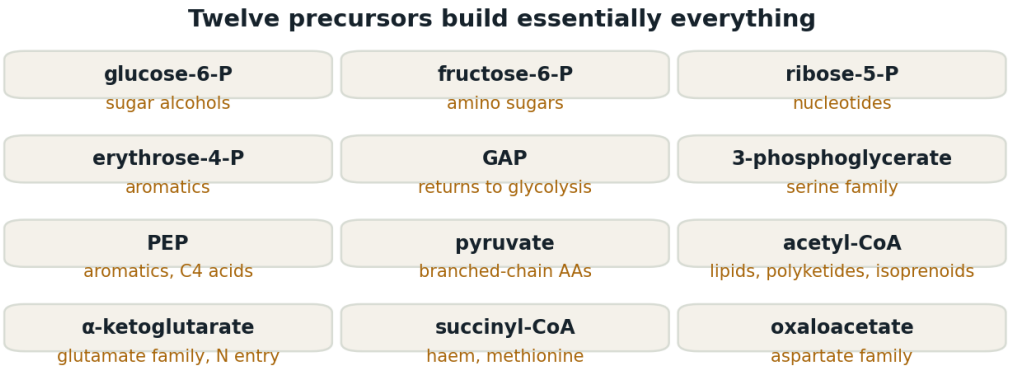
The four families you will actually meet

Fatty acids and polyketides are built by iterative condensation of two-carbon units from acetyl-CoA, or methylmalonyl-CoA for polyketides, with reduction at each cycle. They are therefore acetyl-CoA sinks and NADPH sinks, and the arithmetic in week 58 shows how severe the reductant demand is.

Isoprenoids are built from five-carbon units, either by the mevalonate pathway from acetyl-CoA, which costs ATP and NADPH, or by the methylerythritol phosphate pathway from pyruvate and glyceraldehyde-3-phosphate. Which route your host has, and which you install, changes the balance substantially. Carotenoids, including the astaxanthin you worked on, are in this family.

Amino acids come in families named after their precursor: the aspartate family from oxaloacetate, the glutamate family from α-ketoglutarate, the aromatic family from phosphoenolpyruvate plus erythrose-4-phosphate, and others from pyruvate or 3-phosphoglycerate. Aromatic compounds, including flavonoids and many pharmaceutical intermediates, come through the shikimate pathway from that phosphoenolpyruvate and erythrose-4-phosphate junction, which is why the supply of those two precursors dominates aromatic product engineering.

Nucleotides and their derivatives come from ribose-5-phosphate with amino acid and one-carbon donors. You will meet them less often as products, but their demand is a fixed cost of growth.



The first question about any target molecule is which of these it comes from: that decides what you amplify and what you compete with.

Figure 7.1 The twelve precursors and what each one builds. Use it as a lookup: find your target's precursor and you know which part of metabolism you are about to fight.

Nitrogen, sulfur and one-carbon units

Nitrogen enters metabolism almost entirely through glutamate and glutamine, made from α-ketoglutarate and ammonium. Every amino group in the cell is transferred from there by transamination, which is why α-ketoglutarate availability couples nitrogen assimilation to the TCA cycle. If your product carries nitrogen, your balance needs this.

Sulfur enters through cysteine and methionine, and reducing sulfate to sulfide is expensive in reducing power, a cost often omitted from back-of-envelope balances for sulfur-containing products such as glutathione. One-carbon units move on tetrahydrofolate and on S-adenosylmethionine, which is the methyl donor for nearly every methylation, including the ones that decorate natural products.

The equations from this chapter

Fatty acid cycle	each round: 1 acetyl-CoA (or malonyl-CoA) + 2 NADPH → chain extended by 2 carbons
Nitrogen entry	α-ketoglutarate + NH_4^+ + NAD(P)H → glutamate, then transamination everywhere
Methylation	S-adenosylmethionine is the methyl donor, and it is made from methionine and ATP

Worked example

You want to produce a methylated aromatic compound. Trace the precursor demands and name three things that will limit you.

The aromatic ring comes through the shikimate pathway, which condenses phosphoenolpyruvate and erythrose-4-phosphate, so you need both: phosphoenolpyruvate from glycolysis and erythrose-4-phosphate from the non-oxidative pentose phosphate pathway. The methyl group comes from S-adenosylmethionine, made from methionine and ATP. The three limits, in the order I would check them: erythrose-4-phosphate supply, which is usually the scarcer of the two ring precursors; competition for phosphoenolpyruvate, which is also consumed by the sugar uptake system in many bacteria, so taking it for your product directly impairs glucose uptake; and S-adenosylmethionine regeneration, since methylation leaves a product that must be recycled and the pool is small. A fourth, often decisive, is feedback inhibition on the first shikimate enzyme by the aromatic amino acids.

Three things people get wrong

- ‘Precursor supply is a detail.’ It is usually the binding constraint, and pathway enzymes are rarely the problem.
- ‘Methylation is cheap.’ Each one costs an ATP-derived cofactor that must be regenerated.
- ‘Sulfur is just another element.’ Reducing sulfate costs substantial reducing power.

Self-check — answer on paper first

1. Why does engineering an aromatic product in E. coli often interfere with glucose uptake?

Because the phosphotransferase system that imports glucose consumes phosphoenolpyruvate, which is also one of the two precursors of the shikimate pathway. Diverting phosphoenolpyruvate to the product therefore competes directly with sugar uptake, which is why strains for aromatics often replace the uptake system or re-route it.

2. A target molecule derives from acetyl-CoA and needs many reduction steps. Which two supplies do you check first, and why?

Cytosolic or cellular acetyl-CoA supply, because it is a small, fast-turning pool competed for by the TCA cycle, fatty acids and amino acids; and NADPH, because iterative reduction is the dominant reductant demand and the pentose phosphate pathway pays for it in carbon. Both are supply problems, not pathway-enzyme problems.

3. Where does nitrogen enter metabolism, and what does that couple it to?

Through glutamate and glutamine, formed from α-ketoglutarate and ammonium. That couples nitrogen assimilation to the TCA cycle, since α-ketoglutarate is a cycle intermediate, so nitrogen-rich products draw on the same pool as energy metabolism.

What to open a book for

Lehninger Ch. 20 to 22 covers the chemistry. For the twelve-precursor framing and the yield arithmetic that follows from it, Stephanopoulos Ch. 2 and 3 are better, and that is the book you will keep using.

Molecular biology and gene regulation

This is the week that turns the central dogma from a slogan into a set of parts you can specify, which is what weeks 20 onward will have you assembling.

Transcription, and what a promoter really is

RNA polymerase binds a promoter, melts the duplex, and synthesises RNA from the template strand. In bacteria, promoter recognition is conferred by a sigma factor, an exchangeable subunit that lets one core polymerase read different promoter classes: a housekeeping sigma for most genes, and alternative sigmas for heat shock, stationary phase, nitrogen limitation and so on. That architecture is why a strong promoter in exponential phase can be silent in stationary phase.

A promoter is therefore not simply ‘on’ or ‘off’ but a quantitative part, characterised by its strength, its dependence on sigma factors and on activators and repressors, and its dynamic range from the uninduced to the induced state. When you later choose a promoter from a library, these are the numbers you are choosing among, and week 27 has you measuring them.

Translation and the ribosome binding site

In bacteria the ribosome engages an mRNA at a short sequence upstream of the start codon, and the strength of that interaction, together with the local folding of the mRNA, sets the initiation rate. Because initiation is usually rate-limiting, the ribosome binding site is a second, independent dial on expression. This is practically important: two genes on the same transcript can be expressed at very different levels by changing only their binding sites, which is how you set the stoichiometry of a multi-enzyme pathway without changing its promoter.

A complication to remember: the mRNA sequence immediately after the start codon also folds, so the same binding site can perform differently in front of different genes. This is why predictive calculators for translation initiation are useful but imperfect, and why libraries exist.

Regulation as a circuit

A repressor binds an operator and blocks transcription; an inducer binds the repressor and relieves it. An activator binds and recruits polymerase. Combine them and you have logic: the classic lactose operon is repressed in the absence of lactose and poorly activated in the presence of glucose, so it is expressed well only when lactose is present and glucose is not. Read that as an AND gate implemented in chemistry.

For engineering, three things follow. Inducible systems let you separate growth from production, which matters whenever the product is toxic or the burden is high. Leakiness — expression in the uninduced state — is a real and quantifiable property, and week 28 shows that leak sets a hard ceiling on how far a repressor can turn a gene down. And the inducer itself has a cost and a diffusion problem at scale, which is why density-triggered and self-inducing systems are attractive industrially.

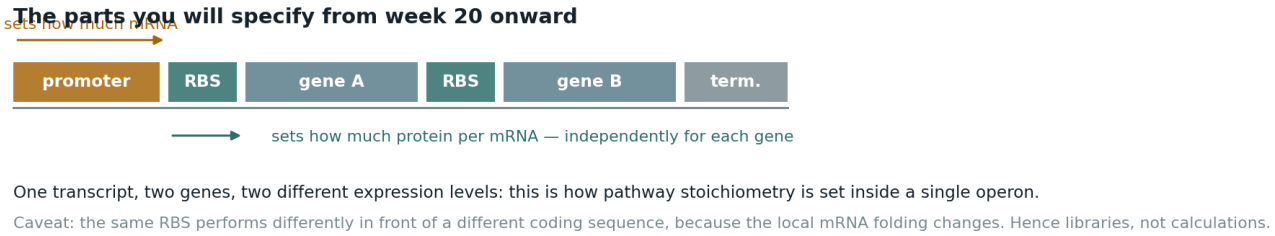


Figure 8.1 The genetic parts you will specify from week 20 onward, and the two independent dials on expression.

Beyond transcription

Expression is also controlled by mRNA stability, by small regulatory RNAs that pair with a transcript and change its translation or degradation, by riboswitches in the untranslated region that change conformation on binding a metabolite, and by protein degradation tags. Each of these is a tool you will use later: small RNAs for tunable knockdown without editing the genome, riboswitches as metabolite sensors, degradation tags for fast clearance. Week 28 treats them as an engineering toolkit.

The equations from this chapter

Expression, loosely	`protein ≈ (transcription rate) × (translation initiation rate) / (degradation + dilution)`
Dilution	`in a growing culture, protein is cleared mainly at the growth rate μ, not by proteolysis`

Worked example

You need three enzymes of a pathway at a ratio of roughly 1 : 5 : 1. Give two ways to achieve this and the trade-off of each.

One: put all three in a single operon under one promoter and set the ratio with ribosome binding sites of different strengths. This keeps the stoichiometry fixed relative to each other whatever the promoter does, which is robust, but it is constrained by context effects — the same binding site performs differently in front of a different coding sequence — so it usually needs a small library and a measurement rather than a calculation. Two: use three separate promoters of different strengths. This is easier to reason about and to change one gene at a time, but the ratio now depends on three promoters that may respond differently to growth phase and conditions, so it can drift. In practice people combine both and screen, which is why week 33 is about plate-based screening rather than calculation.

Three things people get wrong

- ‘The promoter sets expression.’ The promoter and the ribosome binding site each set part of it, independently.
- ‘Uninduced means off.’ Leak is real, measurable, and often the limiting factor.
- ‘A part behaves the same everywhere.’ Context, especially local mRNA folding, changes it.

Self-check — answer on paper first

1. Why can two genes in the same operon be expressed at very different levels?

Because transcription gives them the same mRNA, but translation initiation is set separately for each by its own ribosome binding site and the local folding around its start codon. Differential mRNA stability along the transcript and premature termination also contribute. This is what lets a designer set pathway stoichiometry within one operon.

2. What does a sigma factor do, and why does it matter for a production process?

It confers promoter specificity on RNA polymerase, so which promoters are read depends on which sigma factors are active. It matters because the sigma factor complement changes with growth phase and stress, so a promoter that is strong in exponential growth may be weak in stationary phase or under the stress your product causes, which is precisely when you want production.

3. Why is an inducible system attractive for a toxic product, and what are its two practical drawbacks?

Because it separates a growth phase, in which the pathway is off and the cells accumulate biomass unburdened, from a production phase. The drawbacks are leak, which means the pathway is never fully off and sets a limit on how toxic a product can be tolerated during growth, and the inducer itself, which costs money and must be distributed uniformly in a large vessel, which is why density- or nutrient-triggered systems are preferred at scale.

What to open a book for

Alberts Ch. 4 to 8 selectively, and Lehninger Ch. 24 to 28 for the chemistry of the information pathways. Snyder & Champness is the better book if you want bacterial regulation and genetics specifically, and it is written around experiments rather than mechanisms.

Chapter 9 · tracker week 9

Regulation at the system level: what will fight your design

Your pathway is a tenant in a cell that is optimising for something else. This week is about the landlord.

Carbon catabolite repression

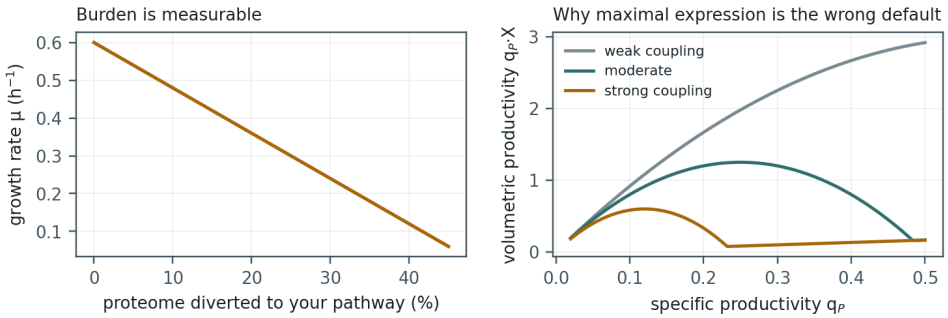
Given two carbon sources, a cell usually consumes the better one first and represses the machinery for the other, producing the familiar two-phase growth curve with a lag between them. In E. coli the mechanism runs through the phosphotransferase system and cyclic AMP; in Bacillus it runs through a different regulator entirely. The details differ by organism, but the principle does not: the cell has a global preference and will enforce it.

Why you care. If your process uses a mixed feedstock — a lignocellulosic hydrolysate with glucose and xylose, for example — catabolite repression means the sugars are consumed sequentially rather than together, which lengthens the fermentation and complicates the feed. Engineering simultaneous utilisation is a recognised and non-trivial objective, and it is a genuine research topic in the Korean groups on your shortlist.

Proteome allocation, and why burden is not a vague complaint

A cell has a finite ribosome capacity and a finite proteome budget. Expressing your pathway means a fraction of that budget is spent on proteins that do nothing for growth, and the cell responds by growing more slowly. This is measurable: the relationship between growth rate and the fraction of proteome allocated to ribosomes, to catabolism and to everything else follows reproducible patterns, and heterologous expression displaces the other sectors.

The engineering consequence is a trade-off you cannot escape by cleverness: more pathway enzyme means less growth, and volumetric productivity depends on both. It is why the right answer is often a two-stage process rather than maximal expression, and why measuring growth rate with and without induction should be among your first experiments.



Illustrative shapes, not measured data: the point is that volumetric productivity is $q_P \cdot X$, so a gain in q_P paid for with biomass can lower output. Two-stage operation is the usual escape.

Figure 9.1 Burden, and why maximal expression is the wrong default. Illustrative shapes rather than measured data: the point is the shape of the trade-off.

Stress responses, and the stringent response

Cells have dedicated programmes for heat, oxidative stress, envelope damage, and nutrient starvation. The one to know by name is the stringent response: on amino acid starvation, an alarmone signal reprogrammes transcription away from ribosomes and growth and towards

survival. Several things you will do — overexpressing a protein that drains an amino acid pool, accumulating a toxic intermediate, running a nutrient-limited fed-batch — can trigger it, and when it fires, production usually falls with growth.

The practical reading of all this: when a strain underperforms, the cause is as likely to be a global regulatory response as a pathway defect. Measuring growth rate, by-product formation and a stress reporter will often tell you more than measuring another pathway intermediate.

The equations from this chapter

Burden	`μ falls roughly in proportion to the proteome fraction diverted to a non-growth protein`
Volumetric productivity	`q_P × X; raising q_P at the expense of X can lower the product`

Worked example

Induction raises your specific productivity by 60% but lowers growth rate from 0.45 to 0.20 h^{-1}. Is induction worth it?

It depends on the process, and the calculation is the answer rather than the biology. Volumetric productivity is specific productivity times biomass, so the question is how much biomass you reach and when. In a batch where induction happens at the start, a growth rate of 0.20 instead of 0.45 means roughly half the doublings in the same time, so you lose far more biomass than the 60% gain in specific rate recovers. In a two-stage process — grow uninduced to the target biomass, then induce — you keep both, which is exactly why decoupled production exists. The general rule: compare on volumetric productivity and on titer at harvest, not on specific rate, and design the staging rather than accepting the trade.

Three things people get wrong

- ‘Burden is hand-waving.’ It is a measurable reduction in growth rate from proteome allocation.
- ‘Mixed sugars are consumed together.’ Usually they are not, and making them so is a project.
- ‘A poor strain means a poor pathway.’ Often it means a global regulatory response you have triggered.

Self-check — answer on paper first

1. Why does catabolite repression matter for a lignocellulosic process?

Because the hydrolysate contains glucose and pentoses, and the cell will consume glucose first and repress pentose utilisation, giving sequential rather than simultaneous consumption. That lengthens the fermentation, leaves residual pentose if the process is stopped early, and complicates feeding, so simultaneous utilisation is an explicit engineering objective.

2. State the trade-off that makes maximal pathway expression the wrong default.

Expressing more pathway protein raises specific productivity but consumes proteome and ribosome capacity, lowering growth rate and therefore biomass. Volumetric productivity is the product of specific rate and biomass, so beyond a point more expression lowers output. The usual resolution is to decouple growth from production in two stages rather than to maximise either alone.

3. A strain underperforms. Name three system-level measurements you would make before touching the pathway.

> Growth rate with and without induction, to quantify burden; by-product formation, especially acetate or ethanol, to see whether flux exceeds respiratory capacity; and substrate consumption with a carbon balance, to find where the carbon actually goes. A stress or stringent-response reporter, and intracellular amino acid or intermediate pools, are reasonable additions.

What to open a book for

No single chapter covers this well. Alberts Ch. 7 for the mechanism of bacterial regulation, then a recent review on proteome allocation and burden, which is where the quantitative account lives. Brock is useful for the physiology.

Chapter 10 · tracker week 10

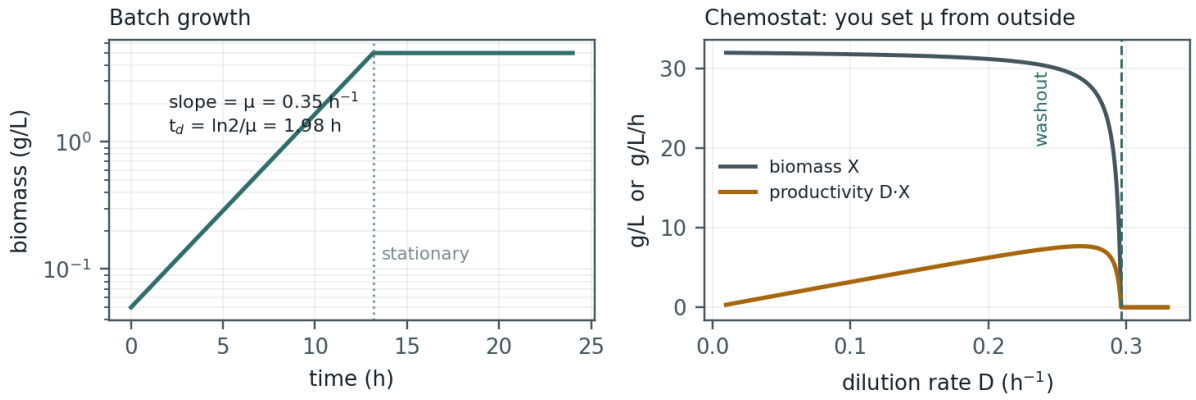
Microbial genetics and microbiology for engineers

The practical biology: how you move DNA, how you select what you want, and how you grow and count cells. This is also the week that ends with the phase 1 checkpoint.

Growth, measured honestly

Batch growth has a lag, an exponential phase in which biomass follows X = X₀·e^(μt), a stationary phase and death. The specific growth rate μ and the doubling time t_d = ln2/μ are the two numbers you quote. In continuous culture at steady state the specific growth rate equals the dilution rate, which is why a chemostat is such a clean instrument: you set the growth rate from outside. Above a critical dilution rate the cells wash out, and week 60 uses that fact.

Three ways of counting disagree, and you should know why. Optical density measures light scattering, which tracks biomass and cell size rather than number, so a culture of filamentous or enlarged cells reads high. Colony counts measure cells able to grow on that medium, so cells that are alive but not culturable are missed. Dry cell weight is the measurement your balances actually need, and it is the one most often skipped. For a yield calculation, use dry weight or a calibration to it, not raw optical density.



Assumed parameters. Note that the most productive dilution rate sits just below the washout cliff, which is an uncomfortable place to run a plant.

Figure 10.1 Batch growth and the chemostat. Note where the most productive dilution rate sits relative to washout.

Getting DNA into cells, and the selection that follows

Three routes matter. Transformation is uptake of free DNA, natural in some species and forced by chemical treatment or electroporation in others. Conjugation is transfer from a donor cell through direct contact, and it is the standard route into organisms such as Streptomyces that resist transformation. Transduction uses a phage to carry DNA. Each has a characteristic efficiency, measured in transformants per microgram, and that efficiency decides whether a library approach is even possible, as the calculation in week 26 shows.

What follows delivery is the part engineers under-appreciate: selection. A selectable marker lets only the cells that took up your DNA grow, usually antibiotic resistance or complementation of an auxotrophy. A counterselectable marker does the opposite: it kills cells that still carry it, which is how you remove a marker after use and how two-step genome editing works. Designing a clean selection is often the difference between a one-week

experiment and a three-month one, and it is why Snyder & Champness is a more useful book here than a molecular biology text.

Mutation, repair and why strains drift

Mutations arise from replication error, chemical damage and mobile elements, and are corrected by several repair systems. Two practical consequences. First, a population is not homogeneous: in a large culture, variants exist, and if a variant grows faster than your producer — typically one that has lost the burden of your pathway — it will take over. That is genetic instability, and it is a routine cause of a strain that performs well at bench scale and fails at production scale, where many more generations elapse. Second, insertion sequences can inactivate your construct, which is why strains with reduced mobile-element content exist.

The defence is design: integrate rather than carry a plasmid where stability matters, avoid unnecessary burden, use a selection that remains in place, and sequence the strain after the fermentation, not only before it.

Sterility, contamination and the things that actually stop work

Aseptic technique is the skill the tracker cannot teach you. The relevant knowledge is: what a sterile medium is and how it is made, why autoclaving and filtration suit different components, what contamination looks like in a plate and in a culture, and that phage contamination in a production plant is a serious event rather than a nuisance. The Bench log tab exists because this is learned at a bench with someone watching, not from a page.

The equations from this chapter

Exponential growth	$X = X_0 \cdot e^{(\mu t)}$; $t_d = \ln 2/\mu$
Chemostat	at steady state $\mu = D$; washout above $\mu_{max} \cdot S_f/(K_s + S_f)$
Library coverage	clones for probability P of seeing a given variant: $n = \ln(1-P)/\ln(1-1/N)$

Worked example

A strain gives 8 g/L at 1 L but 3 g/L in a 1,000 L run from the same inoculum protocol. Give the explanation you would test first, and how.

Genetic instability from the extra generations. Scaling from 1 L to 1,000 L adds roughly ten doublings of inoculum build-up, and if a variant that has lost or silenced the pathway grows even a few per cent faster, it can become a large fraction of the population by the time the production vessel is full. The tests are to plate the final broth and screen colonies for production, to check plasmid retention without selection, and to sequence the construct from the end of the run rather than from the seed. If instability is confirmed, the answers are integration instead of a plasmid, lower burden, or a selection that persists. Alternative explanations worth excluding are oxygen transfer, which falls on scale-up, and gradients in substrate and pH in a large vessel — but instability is the one that matches ‘same protocol, fewer generations, better result’.

Three things people get wrong

- ‘Optical density is biomass.’ It is a correlate; your balances need dry weight.
- ‘A plasmid stays put.’ Without selection and with burden, it does not.
- ‘Scale-up is a process problem.’ Part of it is a genetics problem: more generations, more drift.

Self-check — answer on paper first

1. Why does a chemostat let you set the growth rate, and what is the limit?

Because at steady state the specific growth rate equals the dilution rate: cells must divide exactly as fast as they are removed. The limit is washout, which occurs when the dilution rate exceeds the maximum growth rate achievable on the feed concentration, and the most productive operating point sits just below it, which is an unstable place to run a process.

2. Explain why a clean selection scheme can decide whether an experiment takes a week or three months.

Because the number of cells you must examine depends on whether the desired event is selected or merely screened. A selection lets only the correct cells grow, so a rare event is recovered from a large population in one step. A screen requires you to examine every candidate individually, so the same rare event needs thousands of assays. Designing a selection, including a counterselection for removing markers, is therefore a design task of the same importance as the construct itself.

3. Name the three cell-counting methods and the question each one answers best.

Optical density, for rapidly following the course of a culture; colony counts, for viability in the sense of being able to grow; and dry cell weight, for mass balances and yields. They can disagree legitimately, and quoting a yield from optical density without a calibration to dry weight is a common error.

What to open a book for

Brock is the reference for this entire week, and it is the book most worth having on the shelf for phase 1. MIT OCW 7.13 Experimental Microbial Genetics covers the genetics practically, and Snyder & Champness covers selection design better than anything else.

Before you go on to phase 2

Phase 2 starts from the assumption that you can write a balanced reaction with cofactors, name where a precursor comes from, and say what a promoter and a ribosome binding site each do. If any of those is shaky, the modelling weeks will feel like algebra rather than biology, and the modelling is the part that distinguishes you as a chemical engineer. Go back rather than on.

The phase 1 checkpoint in the tracker's week 10 is the test. Answer it from memory, on paper, before opening the model answers. The three things it asks you to reconstruct — the central carbon map, the global regulators, and a balanced reaction with its cofactors — are exactly the three things weeks 11 to 19 will use on every page.

On the numbers in this module. I checked the quantitative claims against primary compilations rather than from memory, and two were wrong in the first draft: I had given a bacterial cell volume of 2 fL, where measured E. coli values are about 0.4 to 1.8 μm³ and 1 fL is the right round number, and I had narrowed the in-vivo free energy of ATP hydrolysis to 45–55 kJ/mol where the sourced range is about 40 to 60. Both are corrected here, and the cell-volume correction strengthens rather than weakens the conclusion it supports. Where a figure rests on an assumption rather than a measurement, the caption says so. If a number here matters to a calculation you depend on, check it against BioNumbers or the primary source; the mistakes in this document are mine, not a publisher’s.

What this module deliberately leaves out: structural figures, full enzyme mechanisms, and the pathway maps you should annotate yourself. Those are what the books are for, and the chapter notes above say which to open. The module is the spine; it is not a replacement for a library.

Appendix — figures you can legally reuse, and the ones I drew

Every figure in this module I drew myself from the underlying facts. That is legitimate: stoichiometry, pathway topology and rate laws are facts, and facts are not copyrightable — only a particular drawing of them is. What I cannot do is reproduce a figure from Lehninger or Alberts, and I have not.

Three kinds of image are beyond what a drawing can give you: real protein structures, electron and light micrographs of cells, and photographs of equipment and colonies. For those, the sources below are either public domain or openly licensed, so you can download, print and put them in your own notes and even in a thesis, provided you follow the licence. Always check the licence on the individual file rather than trusting the site as a whole, and keep the attribution with the image.

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MIT OpenCourseWare ocw.mit.edu	Lecture notes, slides and video for 7.05, 7.03, 7.13 and 7.344, under a Creative Commons non-commercial share-alike licence. This is the single best free substitute for lectures in phase 1.
KEGG and MetaCyc pathway databases genome.jp/kegg · metacyc.org	Authoritative pathway and reaction data. Use them to check a map and to get reaction stoichiometry. Note that both have licensing restrictions on bulk reuse and on redistribution of their images: read the maps, then draw your own.

BioRender and similar tools free tiers exist	Figure-drawing tools with icon libraries. Convenient, but the free tiers usually restrict publication use, so check the terms before putting an output in a thesis or a paper.

A practical recommendation. For the three gaps in this module, do this: for protein structure, open one enzyme of your own pathway in Mol* on the PDB site and rotate it until the active site makes sense, which teaches more than any printed figure; for cell micrographs, take them from OpenStax or Commons; and for pathway maps, redraw the central carbon map yourself from my Figure 5.1 and check it against KEGG. The redrawing is the part that makes it stick, and it is also the part that makes the figure yours to use.